# What the Low-Resolution Image Still Pins Down

*Interpolation of order 0–5 at 2×, 4× and 8×, and how far an exactly consistent learned decoder goes beyond it*

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| 29.74 → 23.83 dB | +1.09 → +0.51 dB | 25.91 dB | 25.71 dB / 3.5 ms |
|---|---|---|---|
| best interpolation (quintic)<br>2× → 8×, four images | order 0 → 5 gain<br>shrinks from 2× to 8× | best 8× PSNR (assignment)<br>SwinIR-M P1-ft +P | promoted 8× pipeline<br>OURS s7 PSNR-ckpt |

SECTION 1

## The required study: one fixed procedure

Four $512^2$ USC-SIPI images from Assignment 1, picked by a content rule before any score existed: **Mandrill** 4.2.03 (texture), **Jet** 7.1.02 (smooth + sharp edges), **Tank** 7.1.03 (mid-frequency), **Sailboat** 4.2.06 (mixed), as BT.601 luma. Every image is reduced by the course Gaussian pyramid — the one procedure used everywhere in this report:

$$G_{k+1} = \downarrow 2\,(\, g_{\sigma=1} * G_k\,) : 512^2 \to 256^2 \to 128^2 \to 64^2 \qquad (1)$$

$\sigma = 1$ px per ×2 stage, half-pixel sampling, symmetric borders. Each of $256^2$, $128^2$, $64^2$ is enlarged *directly* back to $512^2$ with output pixel u mapped to $x = (u + ½)/s - ½$, so nearest neighbour is exactly the s×s block copy of the prelude. **Methods** (order = kernel degree): nearest (0), bilinear (1), local quadratic and quadratic B-spline (2), Keys cubic $a = -0.5$ and cubic B-spline (3), quartic (4), quintic (5) B-splines, Lanczos-3 (windowed sinc). **Scores:** PSNR ($R = 255$), SSIM (Gaussian $\sigma = 1.5$) on the full clipped $512^2$ frame; **runtime** = median of warm single-thread calls. Results: **Tables 1–3** (interpolation, page 2) and **Table 5** (smoothing choice, page 4).

SECTION 2

## What we built: a decoder that must agree with its input

An interpolated image does not reproduce its own low-resolution input: applying (1) to it gives something else, so part of its error is *visible* in the data. Our decoder removes exactly that part, and learns the rest. (1) is linear, $y = Dx$ with $D(X) = AXA^T$, so with $B = D^T(DD^T)^{-1}$ and $P = I - BD$ any prediction F(y) becomes

$$\hat{x} = F(y) + B(\, y - D\,F(y)\,) = B\,y + P\,F(y) \qquad (2)$$

$$\|\hat{x} - x\|^2 = \|F(y) - x\|^2 - \|(I - P)(F(y) - x)\|^2 \qquad (3)$$

**Technique 1 — learned prior.** F is a 12-layer, 96-channel CNN that works entirely on the small LR grid, ending in one pixel-shuffle added to a bicubic upsampling (0.92–0.97 M weights). It supplies the detail the low-resolution image does not contain. **Technique 2 — exact consistency.** (2) is an orthogonal projection onto the set of images that reduce back to y, so by (3) it can never make the prediction worse, and the output satisfies $D\,\hat{x} = y$ up to float precision. Both run once, in sequence, at decode time; **Table 4** compares them with every alternative.

**CONSISTENCY OF EXACT PROJECTIONS (ALL TEST IMAGES)**

| | |
|---|---|
| max $\lvert D\,\hat{x} - y\rvert$ before clipping | **7.7e−05 grey levels** |
| RMS after clipping to [0, 255] | **0.376 grey levels** |

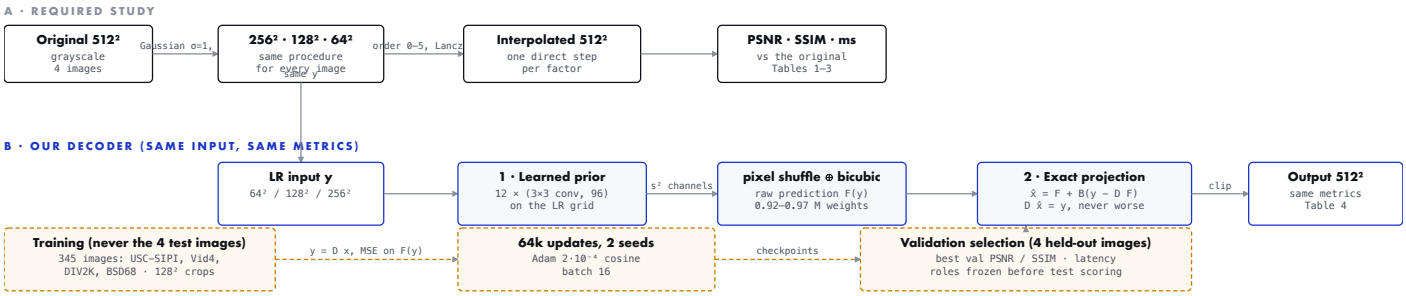



**FIGURE 1 · THE FULL PIPELINE. LANE A** is the required study: one downsampling procedure, then one direct interpolation per factor. **LANE B** is our decoder on the same input y: the learned prior proposes F(y), then the exact projection forces agreement with y before clipping. The dashed path is training — 345 images that exclude all four test images, 64k updates and two seeds per scale, with checkpoints and roles chosen on four held-out validation images and frozen before any test image was scored. The BSD68 part of the pool was kept by a rule fixed in advance (+0.039 dB on validation). References measured in the same harness: every interpolation order ± projection, IBP-20, pretrained SwinIR-M and HAT-L ± projection, and SwinIR-M *fine-tuned on* (1), which separates degradation mismatch from architecture.

**TABLE 4.** Role winners per scale (PSNR dB / SSIM), chosen on validation and latency only; assignment and Urban100-c scored after freezing. Our own family's best 8× checkpoint is OURS s7 PSNR-ckpt +P; DIV2K-20 and per-checkpoint rows in extension/results/role_table.csv.

| SCALE | ROLE | PIPELINE (+P = EXACT PROJECTION) | VALIDATION | MS (× BICUBIC) | ASSIGNMENT 4 | URBAN100-C |
|---|---|---|---|---|---|---|
| 2× | **Highest PSNR** | SwinIR-M P1-ft +P | 34.11 / 0.9182 | 3934.1 (3184×) | 33.03 / 0.886 | 30.90 / 0.923 |
| 2× | **Highest SSIM** | SwinIR-M P1-ft +P | 34.02 / 0.9182 | 3934.1 (3184×) | 33.02 / 0.886 | 30.92 / 0.923 |
| 2× | **Lowest latency** | bicubic (torch) | 29.32 / 0.8338 | 1.2 (1×) | 29.59 / 0.801 | 23.63 / 0.755 |
| 2× | **Best trade-off (promoted)** | bicubic (torch) +P | 31.34 / 0.8961 | 2.1 (2×) | 31.37 / 0.869 | 25.64 / 0.839 |
| 4× | **Highest PSNR** | SwinIR-M P1-ft +P | 29.22 / 0.7765 | 727.7 (681×) | 28.79 / 0.737 | 24.26 / 0.748 |
| 4× | **Highest SSIM** | SwinIR-M P1-ft +P | 29.17 / 0.7768 | 727.7 (681×) | 28.78 / 0.737 | 24.28 / 0.748 |
| 4× | **Lowest latency** | bicubic (torch) | 25.24 / 0.6430 | 1.1 (1×) | 26.08 / 0.640 | 20.22 / 0.528 |
| 4× | **Best trade-off (promoted)** | bicubic (torch) +P | 26.87 / 0.7229 | 1.6 (2×) | 27.60 / 0.709 | 21.54 / 0.614 |
| 8× | **Highest PSNR** | SwinIR-M P1-ft +P | 25.14 / 0.6087 | 387.0 (373×) | 25.91 / 0.611 | 20.64 / 0.544 |
| 8× | **Highest SSIM** | SwinIR-M P1-ft | 25.11 / 0.6090 | 230.9 (223×) | 25.87 / 0.612 | 20.61 / 0.544 |
| 8× | **Lowest latency** | bicubic (torch) | 22.50 / 0.5231 | 1.0 (1×) | 23.76 / 0.550 | 18.13 / 0.395 |
| 8× | **Best trade-off (promoted)** | OURS s7 PSNR-ckpt | 24.77 / 0.5930 | 3.5 (3×) | 25.71 / 0.603 | 19.80 / 0.487 |

# Downscaling inputs, then upscaling scores

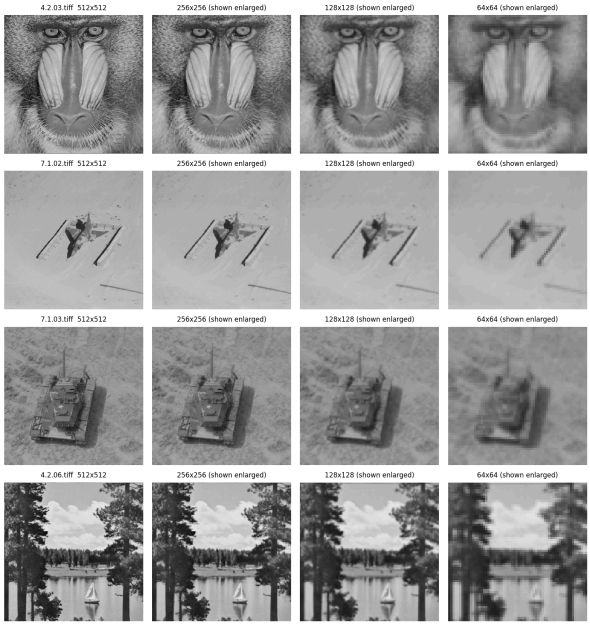



**FIGURE 2 · DOWNSCALING.** The four grayscale originals ($512^2$) and the three downsampled versions of each ($256^2$, $128^2$, $64^2$) produced by (1) — the inputs every method receives. They are shown enlarged with pixel replication so the true sample grid stays visible.

**TABLE 1 · UPSCALING, MEAN OVER THE FOUR IMAGES.** PSNR (dB) / SSIM and median runtime (ms; SciPy for B-splines, NumPy reference for kernel methods). Bold = best interpolation per scale; blue = the two learned decoders of Section 2 on the same inputs (their runtime is the full decode path on the Mac, so it is not comparable with the single-thread interpolation column).

| METHOD (ORDER) | PSNR / SSIM 2× | PSNR / SSIM 4× | PSNR / SSIM 8× | MS 2× | MS 4× | MS 8× |
|---|---|---|---|---|---|---|
| Nearest (0) | 28.65 / 0.783 | 25.42 / 0.615 | 23.32 / 0.525 | 3.65 | 3.61 | 3.80 |
| Bilinear (1) | 28.64 / 0.765 | 25.53 / 0.620 | 23.39 / 0.542 | 5.49 | 5.48 | 5.70 |
| Quadratic local (2) | 29.23 / 0.790 | 25.80 / 0.632 | 23.57 / 0.547 | 2.28 | 1.80 | 1.53 |
| Quadratic B-spline (2) | 29.58 / 0.802 | 26.06 / 0.641 | 23.75 / 0.551 | 9.55 | 8.74 | 8.92 |
| Bicubic Keys a=-0.5 (3) | 29.37 / 0.794 | 25.95 / 0.636 | 23.68 / 0.549 | 2.17 | 1.84 | 1.69 |
| Cubic B-spline (3) | 29.62 / 0.802 | 26.10 / 0.642 | 23.78 / 0.552 | 14.37 | 13.63 | 14.38 |
| Quartic B-spline (4) | 29.71 / 0.806 | 26.14 / 0.644 | 23.81 / 0.553 | 21.53 | 20.67 | 20.69 |
| Quintic B-spline (5) | **29.74 / 0.807** | **26.16 / 0.645** | **23.83 / 0.553** | 27.74 | 27.12 | 27.23 |
| Lanczos-3 (sinc ref.) | 29.70 / 0.805 | 26.15 / 0.643 | 23.81 / 0.552 | 3.60 | 2.99 | 2.62 |
| **EXTENSION ON THE SAME IMAGES (NOT INTERPOLATION)** | RECONSTRUCTION BACK TO $512^2$ FROM THE SAME LOW-RESOLUTION INPUTS | | | | | |
| **OURS · DirectLR w96 d12 + exact projection** | **32.64 / 0.881** | **28.39 / 0.728** | **25.72 / 0.603** | **40.7** | **14.1** | **7.4** |
| **SwinIR-M fine-tuned on (1) + exact projection** | **33.03 / 0.886** | **28.79 / 0.737** | **25.91 / 0.611** | **3934.1** | **727.7** | **387.0** |

**TABLE 2 · UPSCALING.** PSNR (dB) for every image, interpolation method and factor. Bold = best for that image and factor.

| METHOD | 2× ($256^2 \to 512^2$) | | | | 4× ($128^2 \to 512^2$) | | | | 8× ($64^2 \to 512^2$) | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | MANDRILL | JET | TANK | SAILBOAT | MANDRILL | JET | TANK | SAILBOAT | MANDRILL | JET | TANK | SAILBOAT |
| Nearest (0) | 22.59 | 33.63 | 31.40 | 26.96 | 20.46 | 29.86 | 28.49 | 22.89 | 19.47 | 27.17 | 26.58 | 20.05 |
| Bilinear (1) | 22.31 | 33.80 | 31.25 | 27.18 | 20.44 | 30.02 | 28.56 | 23.08 | 19.50 | 27.18 | 26.68 | 20.18 |
| Quadratic local (2) | 22.73 | 34.50 | 31.77 | 27.94 | 20.56 | 30.41 | 28.79 | 23.45 | 19.57 | 27.43 | 26.85 | 20.43 |
| Quadratic B-spline (2) | 22.95 | 34.90 | 32.05 | 28.43 | 20.66 | 30.79 | 28.99 | 23.79 | 19.64 | 27.69 | 27.01 | 20.67 |
| Bicubic Keys a=-0.5 (3) | 22.80 | 34.68 | 31.87 | 28.15 | 20.62 | 30.64 | 28.90 | 23.65 | 19.61 | 27.58 | 26.94 | 20.57 |
| Cubic B-spline (3) | 22.97 | 34.94 | 32.08 | 28.51 | 20.68 | 30.85 | 29.01 | 23.84 | 19.66 | 27.73 | 27.03 | 20.71 |
| Quartic B-spline (4) | 23.04 | 35.02 | 32.16 | 28.63 | 20.70 | 30.91 | 29.05 | 23.91 | 19.67 | 27.77 | 27.06 | 20.75 |
| Quintic B-spline (5) | **23.06** | **35.05** | **32.18** | **28.67** | **20.71** | **30.94** | **29.07** | **23.93** | **19.68** | **27.79** | **27.07** | **20.77** |
| Lanczos-3 (sinc ref.) | 23.02 | 35.03 | 32.14 | 28.62 | 20.70 | 30.93 | 29.05 | 23.91 | 19.67 | 27.78 | 27.06 | 20.75 |

**TABLE 3 · UPSCALING.** SSIM for every image, interpolation method and factor.

| METHOD | 2× ($256^2 \to 512^2$) | | | | 4× ($128^2 \to 512^2$) | | | | 8× ($64^2 \to 512^2$) | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | MANDRILL | JET | TANK | SAILBOAT | MANDRILL | JET | TANK | SAILBOAT | MANDRILL | JET | TANK | SAILBOAT |
| Nearest (0) | 0.629 | 0.917 | 0.777 | 0.811 | 0.368 | 0.859 | 0.612 | 0.622 | 0.266 | 0.825 | 0.529 | 0.481 |
| Bilinear (1) | 0.582 | 0.913 | 0.760 | 0.805 | 0.362 | 0.863 | 0.617 | 0.639 | 0.276 | 0.834 | 0.545 | 0.514 |
| Quadratic local (2) | 0.627 | 0.921 | 0.785 | 0.827 | 0.377 | 0.867 | 0.628 | 0.655 | 0.281 | 0.835 | 0.549 | 0.522 |
| Quadratic B-spline (2) | 0.647 | 0.925 | 0.796 | 0.839 | 0.387 | 0.871 | 0.636 | 0.669 | 0.285 | 0.837 | 0.553 | 0.530 |
| Bicubic Keys a=-0.5 (3) | 0.632 | 0.922 | 0.788 | 0.832 | 0.382 | 0.869 | 0.632 | 0.662 | 0.283 | 0.836 | 0.551 | 0.526 |
| Cubic B-spline (3) | 0.648 | 0.925 | 0.797 | 0.840 | 0.388 | 0.871 | 0.637 | 0.671 | 0.285 | 0.837 | 0.554 | 0.531 |
| Quartic B-spline (4) | 0.654 | 0.926 | 0.801 | 0.843 | 0.391 | 0.872 | 0.639 | 0.674 | 0.286 | 0.838 | 0.554 | 0.532 |
| Quintic B-spline (5) | **0.656** | **0.927** | **0.802** | **0.843** | **0.392** | **0.872** | **0.640** | **0.675** | **0.286** | **0.838** | **0.555** | **0.533** |
| Lanczos-3 (sinc ref.) | 0.653 | 0.926 | 0.800 | 0.842 | 0.390 | 0.872 | 0.639 | 0.673 | 0.286 | 0.837 | 0.554 | 0.531 |

# Upscaled results: reconstructions and artefacts

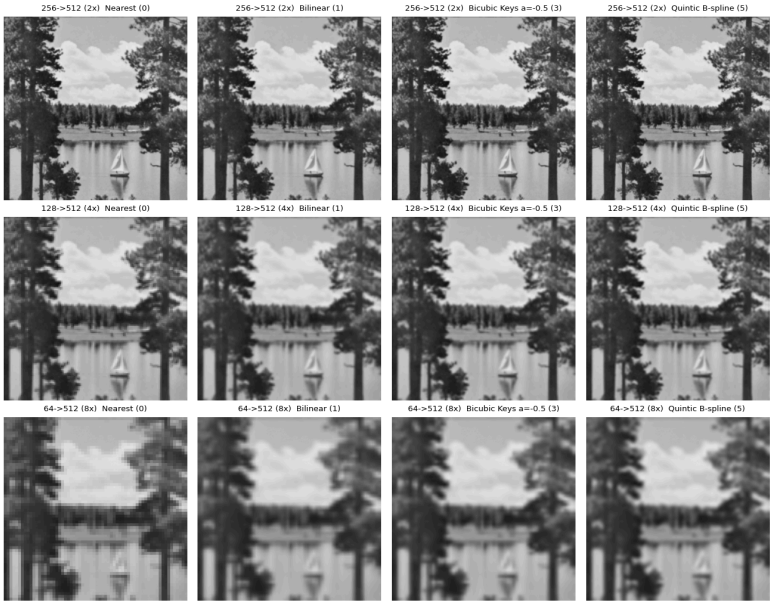



**FIGURE 3.** Sailboat enlarged from $256^2$, $128^2$ and $64^2$ (rows) with nearest, bilinear, Keys bicubic and quintic B-spline (columns).

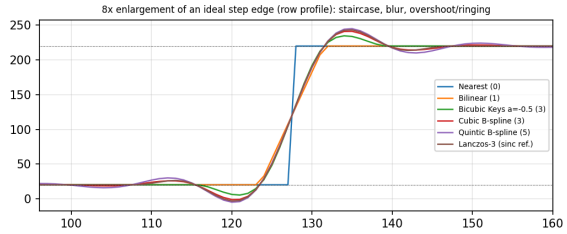



**FIGURE 4.** An ideal step enlarged 8×: nearest = staircase, bilinear = ramp (blur), order ≥ 3 = overshoot and ringing on both sides.

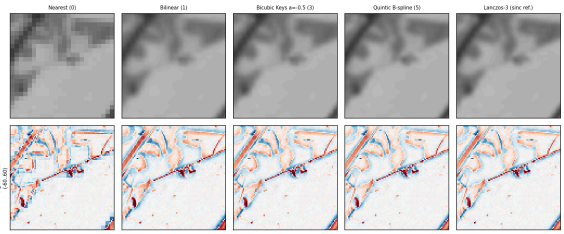


**FIGURE 5.** 4× Jet crop and signed error (red = too bright): every kernel's error sits on edges.

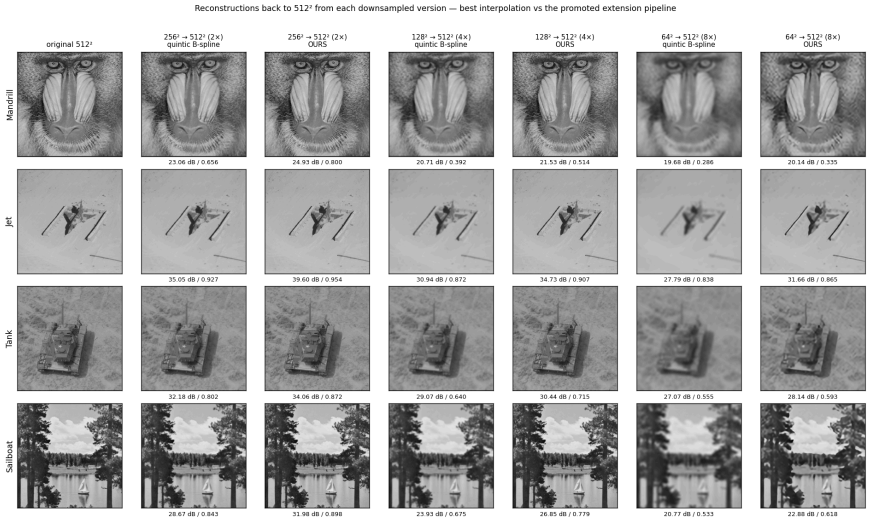



**FIGURE 6 · THE RECONSTRUCTIONS.** Every test image restored to $512^2$ from each of its three downsampled versions: the best interpolation (quintic B-spline) beside the promoted extension pipeline of Table 4, with full-image PSNR / SSIM under each panel. Crops of all four images for seven kernels at every factor are in figures/fig2_sr_crops_*.png.

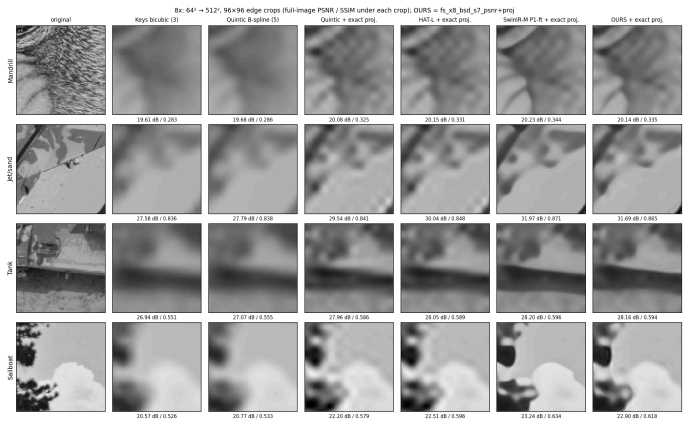



**FIGURE 7.** 8× edge crops of all four images: interpolation, projection, pretrained HAT-L, SwinIR-M fine-tuned on (1) and OURS (full-image PSNR / SSIM under each crop). Crops for the other images and factors: figures/ and extension/figures/.

# Discussion, best choices and why

SECTION 3

## How quality changes from 2× to 8×

Quality falls steeply with the factor: quintic gives 29.74 → 26.16 → 23.83 dB and SSIM 0.807 → 0.553. A $64^2$ input holds 1.56% of the samples, and the pyramid already removed detail finer than its grid. The gap between methods shrinks with it: order 0 → 5 gains 1.09 / 0.74 / 0.51 dB. *Why:* the error splits into energy above the LR Nyquist limit, which no interpolator can recover, plus a passband term the kernel controls; B-splines approach the ideal low-pass, so their gains saturate while the lost band grows with s.

SECTION 4

## Visual differences and artefacts

- **Nearest (0)**: blocking and staircase edges (Figs. 2–3), no overshoot; it keeps local contrast, so its SSIM beats bilinear at 2× (0.783 vs 0.765).
- **Bilinear (1)**: no blocks but the strongest blur — its triangle kernel is a second low-pass on top of the pyramid.
- **Order 2**: the local quadratic (29.23 dB) is below the quadratic B-spline (29.58 dB); "second order" must name its kernel.
- **Order ≥ 3**: sharper edges with overshoot/ringing from negative lobes; mean local overshoot at 8× rises Keys 0.031 → cubic 0.057 → quintic 0.074 grey levels.

SECTION 5

## Quality versus runtime, and content

Quality rises with spline order with diminishing returns — cubic → quintic adds 0.12 dB at 2× and 0.05 dB at 8× while SciPy time about doubles (14 → 28 ms). A cubic kernel gives the best quality per millisecond. **Smooth regions** are easy for every method (> 33.7 dB at 8×); **edges** are hardest and gain most from order (+1.7 dB vs +0.2 dB smooth, 2×); **texture** converges: Mandrill peaks at 23.06 dB even at 2× (Jet 35.05 dB), and there nearest (22.59) even beats bilinear (22.31).

SECTION 6

## Best smoothing, and the best pipeline overall

The required tables keep one fixed downsampler. A separate question is which smoothing, and which *pair* of downsampler and interpolator, reconstructs best (same four images, each level enlarged directly to $512^2$). Lighter anti-aliasing wins for fidelity: the pyramid's Gaussian keeps 82% of the passband at 8× (gain 0.27 at 0.9 $f_c$) against Lanczos-3's 98% (0.68), both with ≤ 0.2% alias energy; no smoothing aliases badly (22.71 dB at 8×). Recursive and direct smoothing agree to ≤ 0.02 dB.

**TABLE 5.** Downsampling comparison: mean PSNR (dB) of the best interpolator for each smoothing (its name in grey; SSIM in results/downsampler_joint.csv).

| DOWNSAMPLING | 2× | 4× | 8× |
|---|---|---|---|
| no smoothing (decimation) | 29.79 Keys | 25.69 Keys | 22.94 Keys |
| Gaussian σ=1.4 | 28.30 cubic | 25.10 cubic | 23.03 cubic |
| Gaussian σ=1 (P1, frozen) | 29.74 quintic | 26.16 quintic | 23.83 quintic |
| Gaussian σ=0.5 | 30.84 cubic | 27.16 cubic | 24.59 cubic |
| box / area | 30.87 cubic | 27.19 cubic | 24.61 cubic |
| Lanczos-3 direct | 31.30 quintic | 27.50 quintic | 24.82 quintic |
| DCT band crop (ideal low-pass) | 31.74 DCT | 27.73 DCT | 24.96 DCT |
| co-designed for the upscaler (8-bit LR) | 31.62 quintic | 27.70 quintic | 24.97 quintic |

**Best pipeline, end to end.** With the downsampler fixed by the task, the best decoders are those of Table 4. If both ends may be designed (a codec), the best measured pair is **DCT band crop (ideal low-pass) + DCT zero-pad** at 2× (31.74 dB) and **co-designed for the upscaler (float LR) + quintic B-spline** at 8× (24.97 dB), i.e. +2.00 and +1.15 dB over the frozen Gaussian with the same interpolator.

SECTION 7

## Beyond interpolation: who wins and why

- **2×**: PSNR — SwinIR-M P1-ft +P; SSIM — SwinIR-M P1-ft +P; fastest — bicubic (torch); **promoted — bicubic (torch) +P** (31.37 dB / 0.869 on the assignment images, 2.1 ms).
- **4×**: PSNR — SwinIR-M P1-ft +P; SSIM — SwinIR-M P1-ft +P; fastest — bicubic (torch); **promoted — bicubic (torch) +P** (27.60 dB / 0.709 on the assignment images, 1.6 ms).
- **8×**: PSNR — SwinIR-M P1-ft +P; SSIM — SwinIR-M P1-ft; fastest — bicubic (torch); **promoted — OURS s7 PSNR-ckpt** (25.71 dB / 0.603 on the assignment images, 3.5 ms).

- **Projection never hurts** (clean y): (3) is an orthogonal projection onto {z : Dz = y}, which contains x. Measured: projected ≥ raw PSNR in 5576/5576 image–pipeline pairs.
- **Learning beats the linear ceiling**: the best consistent linear method is an oblique projection capped at 24.97 dB (8×, assignment images; bound verified 108/108); the MSE-optimal E[x | y] is non-linear, and P F(y) learns only the missing null-space part.
- **Degradation must match**: networks trained on bicubic inputs estimate $E[x \mid y_{\text{bicubic}}]$; fine-tuning SwinIR-M on (1) removes that bias, and it leads on quality at every scale (Urban100-c: 2× +2.13 dB over HAT-L + P, 4× +1.83 dB over HAT-L + P, 8× +1.02 dB over HAT-L + P).
- **Learning helps most where the input still holds information**: OURS over quintic + projection on Urban100-c is +3.01 / +1.39 / +0.66 dB at 2× / 4× / 8×; over pretrained HAT-L + projection -0.04 / +0.52 / +0.20 dB, while running 449× / 474× / 3044× faster on the Mac.
- **The trade-off winner changes with scale**: convolution cost grows with the LR area $(512/s)^2$, ×4 per halving of s — OURS takes 3.5 / 11.1 / 42.6 ms at 8× / 4× / 2×, while bicubic stays ≈ 1 ms, so the 10×-bicubic budget admits the network only at large factors.

**LIMITS OF THE CLAIMS**

One noise-free synthetic degradation and one machine (HAT-L needs tiling on 8 GB, so its time is machine-specific). The assignment images were viewed during development, so their extension scores are descriptive; Urban100-c and DIV2K-20 were never trained on, but pretrained networks may have seen them upstream. Validation scores of selected checkpoints are optimistic. Claim: best measured choices among the evaluated pipelines for this downsampler on this hardware — not a general state-of-the-art claim. Paired bootstrap statistics, seeds, latency records and the coverage ledger: extension/results/.

SECTION 8

## Conclusion

**Interpolation:** quality rises with order (quintic ≈ Lanczos > cubic > quadratic > bilinear ≈ nearest) but the gains are small beside the loss from larger factors and from content; a cubic kernel is the best quality per unit time. **Beyond it:** exact consistency with the observed image is the cheapest large gain (bicubic + projection is the promoted pipeline at 2× and 4×); the best quality at every scale comes from SwinIR-M fine-tuned on this downsampler; and OURS is the promoted pipeline at 8×, the only learned model inside the 10×-bicubic budget there.

[1] Keys, IEEE TASSP 1981 · [2] Unser, IEEE SPM 1999 · [3] Getreuer, IPOL 2011 · [4] Burt & Adelson, IEEE TCOM 1983 · [5] Irani & Peleg, CVGIP 1991 · [6] Schwab, Antholzer & Haltmeier, Inverse Problems 2019 · [7] Bahat & Michaeli, CVPR 2020 · [8] Wang, Yu & Zhang (DDNM), ICLR 2023 · [9] Shi et al. (ESPCN), CVPR 2016 · [10] Liang et al. (SwinIR), ICCVW 2021 · [11] Chen et al. (HAT), CVPR 2023 · [12] Wang et al. (SSIM), IEEE TIP 2004 · [13] Blau & Michaeli, CVPR 2018 · Data: USC-SIPI, DIV2K, BSD68, Urban100, Vid4.